MSSV: 23521662

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BÀI THỰC HÀNH 01

Bài 1:

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hadoop@trinh23521662-VirtualBox:~/lab1$ hdfs dfs -cat /lab1/output/1/part-r-00000
 American Beauty (1999) Average rating: 3.75 (Total ratings: 2)
 Avatar (2009)  Average rating: 4.75 (Total ratings: 2)
 Back to the Future (1985)      Average rating: 4.0 (Total ratings: 2)
 Birdman (2014) Average rating: 3.0 (Total ratings: 2)
 Braveheart (1995)      Average rating: 4.25 (Total ratings: 2)
 Coco (2017)    Average rating: 4.25 (Total ratings: 2)
 Dunkirk (2017) Average rating: 3.5 (Total ratings: 2)
 Fight Club (1999)      Average rating: 4.25 (Total ratings: 2)
 Forrest Gump (1994)    Average rating: 3.75 (Total ratings: 2)
 Gladiator (2000)       Average rating: 3.25 (Total ratings: 2)
 Good Will Hunting (1997)       Average rating: 3.5 (Total ratings: 2)
 Gravity (2013) Average rating: 4.25 (Total ratings: 2)
 Inception (2010)       Average rating: 3.75 (Total ratings: 2)
 Interstellar (2014)    Average rating: 3.5 (Total ratings: 2)
 Jurassic Park (1993)   Average rating: 4.0 (Total ratings: 2)
 La La Land (2016)      Average rating: 4.0 (Total ratings: 2)
 Mad Max: Fury Road (2015)      Average rating: 4.0 (Total ratings: 2)
 Manchester by the Sea (2016)   Average rating: 4.0 (Total ratings: 2)
 Memento (2000) Average rating: 4.0 (Total ratings: 2)
 Moonlight (2016)       Average rating: 3.5 (Total ratings: 2)
 Parasite (2019)        Average rating: 4.0 (Total ratings: 2)
 Pulp Fiction (1994)    Average rating: 4.0 (Total ratings: 2)
 Saving Private Ryan (1998)     Average rating: 4.0 (Total ratings: 2)
 Schindler's List (1993)        Average rating: 4.25 (Total ratings: 2)
 Se7en (1995)   Average rating: 3.75 (Total ratings: 2)
 Slumdog Millionaire (2008)     Average rating: 4.25 (Total ratings: 2)
 Spotlight (2015)       Average rating: 4.25 (Total ratings: 2)
 Star Wars: A New Hope (1977)   Average rating: 3.5 (Total ratings: 2)
 The Avengers (2012)    Average rating: 3.75 (Total ratings: 2)
 The Big Short (2015)   Average rating: 4.5 (Total ratings: 2)
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 The Big Short (2015)   Average rating: 4.5 (Total ratings: 2)
 The Dark Knight (2008) Average rating: 3.5 (Total ratings: 2)
 The Departed (2006)    Average rating: 3.25 (Total ratings: 2)
 The Godfather (1972)   Average rating: 3.5 (Total ratings: 2)
 The Green Mile (1999)  Average rating: 3.5 (Total ratings: 2)
 The Imitation Game (2014)      Average rating: 3.75 (Total ratings: 2)
 The Lion King (1994)   Average rating: 4.0 (Total ratings: 2)
 The Lord of the Rings: The Fellowship of the Ring (2001)Average rating: 4.25 (Total ratings: 2)
 The Lord of the Rings: The Return of the King (2003)   Average rating: 3.5 (Total ratings: 2)
 The Lord of the Rings: The Two Towers (2002)   Average rating: 3.75 (Total ratings: 2)
 The Matrix (1999)      Average rating: 3.75 (Total ratings: 2)
 The Prestige (2006)    Average rating: 4.0 (Total ratings: 2)
 The Revenant (2015)    Average rating: 3.5 (Total ratings: 2)
 The Shawshank Redemption (1994)        Average rating: 4.0 (Total ratings: 2)
 The Silence of the Lambs (1991)        Average rating: 3.5 (Total ratings: 2)
 The Social Network (2010)      Average rating: 3.75 (Total ratings: 2)
 The Usual Suspects (1995)      Average rating: 4.25 (Total ratings: 2)
 The Wolf of Wall Street (2013) Average rating: 4.25 (Total ratings: 2)
 Titanic (1997) Average rating: 3.5 (Total ratings: 2)
 Toy Story (1995)       Average rating: 4.25 (Total ratings: 2)
 Whiplash (2014)        Average rating: 3.75 (Total ratings: 2)
RESULT  No movies has at least 5 ratings.
hadoop@trinh23521662-VirtualBox:~/lab1$
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Bài 2:

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hadoop@trinh23521662-VirtualBox:~/lab1$ hdfs dfs -cat /lab1/output2/part-r-00000
Action  Avg: 3.73,    Count: 20
Adventure       Avg: 3.85,    Count: 30
Animation       Avg: 4.17,    Count: 6
Biography       Avg: 4.14,    Count: 14
Children        Avg: 4.25,    Count: 2
Comedy  Avg: 3.96,    Count: 12
Crime   Avg: 3.78,    Count: 20
Drama   Avg: 3.81,    Count: 76
Family  Avg: 4.25,    Count: 2
Fantasy Avg: 3.95,    Count: 10
History Avg: 4.00,    Count: 6
Music   Avg: 3.88,    Count: 4
Mystery Avg: 4.00,    Count: 8
Romance Avg: 3.75,    Count: 8
Sci-Fi  Avg: 3.85,    Count: 20
Thriller        Avg: 3.88,    Count: 16
War     Avg: 4.00,    Count: 2
hadoop@trinh23521662-VirtualBox:~/lab1$
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Bài 3:

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hadoop@trinh23521662-VirtualBox:~/lab1$ hdfs dfs -cat /lab1/output/3/part-r-00000
American Beauty (1999)  Male: 3.00, Female: 4.50
Avatar (2009)   Male: 5.00, Female: 4.50
Back to the Future (1985)       Male: 4.00, Female: 4.00
Birdman (2014)  Male: 3.00, Female: 3.00
Braveheart (1995)       Male: 4.50, Female: 4.00
Coco (2017)     Male: 5.00, Female: 3.50
Dunkirk (2017)  Male: 4.00, Female: 3.00
Fight Club (1999)       Male: 5.00, Female: 3.50
Forrest Gump (1994)     Male: 2.50, Female: 5.00
Gladiator (2000)        Male: 3.50, Female: 3.00
Good Will Hunting (1997)        Male: 3.00, Female: 4.00
Gravity (2013)  Male: 4.00, Female: 4.50
Inception (2010)        Male: 3.50, Female: 4.00
Interstellar (2014)     Male: 3.50, Female: 3.50
Jurassic Park (1993)    Male: 3.50, Female: 4.50
La La Land (2016)       Male: 5.00, Female: 3.00
Mad Max: Fury Road (2015)       Male: 4.00, Female: 4.00
Manchester by the Sea (2016)    Male: 3.50, Female: 4.50
Memento (2000)  Male: 3.50, Female: 4.50
Moonlight (2016)        Male: 3.00, Female: 4.00
Parasite (2019) Male: 4.00, Female: 4.00
Pulp Fiction (1994)     Male: 4.50, Female: 3.50
Saving Private Ryan (1998)      Male: 4.00, Female: 4.00
Schindler's List (1993) Male: 4.00, Female: 4.50
Se7en (1995)    Male: 4.00, Female: 3.50
Slumdog Millionaire (2008)      Male: 4.50, Female: 4.00
Spotlight (2015)        Male: 4.50, Female: 4.00
Star Wars: A New Hope (1977)    Male: 4.00, Female: 3.00
The Avengers (2012)     Male: 3.50, Female: 4.00
The Big Short (2015)    Male: 5.00, Female: 4.00
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The Big Short (2015)    Male: 5.00, Female: 4.00
The Dark Knight (2008)  Male: 4.00, Female: 3.00
The Departed (2006)     Male: 4.50, Female: 2.00
The Godfather (1972)    Male: 4.50, Female: 2.50
The Green Mile (1999)   Male: 3.00, Female: 4.00
The Imitation Game (2014)       Male: 4.50, Female: 3.00
The Lion King (1994)    Male: 4.50, Female: 3.50
The Lord of the Rings: The Fellowship of the Ring (2001) Male: 4.50, Female: 4.00
The Lord of the Rings: The Return of the King (2003)    Male: 4.00, Female: 3.00
The Lord of the Rings: The Two Towers (2002)    Male: 4.50, Female: 3.00
The Matrix (1999)       Male: 3.00, Female: 4.50
The Prestige (2006)     Male: 4.00, Female: 4.00
The Revenant (2015)     Male: 3.50, Female: 3.50
The Shawshank Redemption (1994) Male: 3.00, Female: 5.00
The Silence of the Lambs (1991) Male: 3.50, Female: 3.50
The Social Network (2010)       Male: 3.50, Female: 4.00
The Usual Suspects (1995)       Male: 4.00, Female: 4.50
The Wolf of Wall Street (2013)  Male: 4.00, Female: 4.50
Titanic (1997)  Male: 4.00, Female: 3.00
Toy Story (1995)        Male: 4.50, Female: 4.00
Whiplash (2014) Male: 4.50, Female: 3.00
hadoop@trinh23521662-VirtualBox:~/lab1$
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Bài 4:

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hadoop@trinh23521662-VirtualBox:~/lab1$ hdfs dfs -cat /lab1/output/4/part-r-00000
American Beauty (1999)  0-18: NA        18-35: 3.7535-50: NA       50+: NA
Avatar (2009)   0-18: NA        18-35: 4.50     35-50: 5.00        50+: NA
Back to the Future (1985)       0-18: NA        18-35: 4.00        35-50: NA       50+: NA
Birdman (2014)  0-18: NA        18-35: 3.00     35-50: NA  50+: NA
Braveheart (1995)       0-18: NA        18-35: 4.0035-50: 4.50     50+: NA
Coco (2017)     0-18: NA        18-35: 4.25     35-50: NA  50+: NA
Dunkirk (2017)  0-18: NA        18-35: 3.50     35-50: NA  50+: NA
Fight Club (1999)       0-18: NA        18-35: 4.2535-50: NA       50+: NA
Forrest Gump (1994)     0-18: NA        18-35: 3.7535-50: NA       50+: NA
Gladiator (2000)        0-18: NA        18-35: 3.0035-50: 3.50     50+: NA
Good Will Hunting (1997)        0-18: NA        18-35: 3.50        35-50: NA       50+: NA
Gravity (2013)  0-18: NA        18-35: 4.50     35-50: 4.00        50+: NA
Inception (2010)        0-18: NA        18-35: 3.7535-50: NA       50+: NA
Interstellar (2014)     0-18: NA        18-35: 3.5035-50: 3.50     50+: NA
Jurassic Park (1993)    0-18: NA        18-35: 4.0035-50: NA       50+: NA
La La Land (2016)       0-18: NA        18-35: 3.0035-50: 5.00     50+: NA
Mad Max: Fury Road (2015)       0-18: NA        18-35: 4.00        35-50: NA       50+: NA
Manchester by the Sea (2016)    0-18: NA        18-35: 4.50        35-50: 3.50     50+: NA
Memento (2000)  0-18: NA        18-35: 4.50     35-50: 3.50        50+: NA
Moonlight (2016)        0-18: NA        18-35: 4.0035-50: 3.00     50+: NA
Parasite (2019) 0-18: NA        18-35: 4.00     35-50: 4.00        50+: NA
Pulp Fiction (1994)     0-18: NA        18-35: 4.0035-50: NA       50+: NA
Saving Private Ryan (1998)      0-18: NA        18-35: 4.00        35-50: 4.00     50+: NA
Schindler's List (1993) 0-18: NA        18-35: 4.5035-50: 4.00     50+: NA
Se7en (1995)    0-18: NA        18-35: 3.75     35-50: NA  50+: NA
Slumdog Millionaire (2008)      0-18: NA        18-35: 4.00        35-50: 4.50     50+: NA
Spotlight (2015)        0-18: NA        18-35: 4.0035-50: 4.50     50+: NA
Star Wars: A New Hope (1977)    0-18: NA        18-35: 3.00        35-50: 4.00     50+: NA
The Avengers (2012)     0-18: NA        18-35: 3.7535-50: NA       50+: NA
The Big Short (2015)    0-18: NA        18-35: 4.5035-50: NA       50+: NA
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The Big Short (2015)    0-18: NA        18-35: 4.5035-50: NA       50+: NA
The Dark Knight (2008)  0-18: NA        18-35: 3.0035-50: 4.00     50+: NA
The Departed (2006)     0-18: NA        18-35: 2.0035-50: 4.50     50+: NA
The Godfather (1972)    0-18: NA        18-35: 3.5035-50: NA       50+: NA
The Green Mile (1999)   0-18: NA        18-35: 4.0035-50: 3.00     50+: NA
The Imitation Game (2014)       0-18: NA        18-35: 3.00        35-50: 4.50     50+: NA
The Lion King (1994)    0-18: NA        18-35: 4.0035-50: NA       50+: NA
The Lord of the Rings: The Fellowship of the Ring (2001)   0-18: NA        18-35: 4.00     35-50: 4.5050+: NA
The Lord of the Rings: The Return of the King (2003)       0-18: NA        18-35: 3.50     35-50: NA 50+: NA
The Lord of the Rings: The Two Towers (2002)    0-18: NA   18-35: 3.00     35-50: 4.50     50+: NA
The Matrix (1999)       0-18: NA        18-35: 3.7535-50: NA       50+: NA
The Prestige (2006)     0-18: NA        18-35: 4.0035-50: 4.00     50+: NA
The Revenant (2015)     0-18: NA        18-35: 3.5035-50: NA       50+: NA
The Shawshank Redemption (1994) 0-18: NA        18-35: 4.00        35-50: NA       50+: NA
The Silence of the Lambs (1991) 0-18: NA        18-35: 3.50        35-50: 3.50     50+: NA
The Social Network (2010)       0-18: NA        18-35: 4.00        35-50: 3.50     50+: NA
The Usual Suspects (1995)       0-18: NA        18-35: 4.25        35-50: NA       50+: NA
The Wolf of Wall Street (2013)  0-18: NA        18-35: 4.50        35-50: 4.00     50+: NA
Titanic (1997)  0-18: NA        18-35: 3.00     35-50: 4.00        50+: NA
Toy Story (1995)        0-18: NA        18-35: 4.0035-50: 4.50     50+: NA
Whiplash (2014) 0-18: NA        18-35: 3.00     35-50: 4.50        50+: NA
hadoop@trinh23521662-VirtualBox:~/lab1$
```